AI Multilingual Chatbot for Lung Cancer Detection

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***Abstract*—Lung cancer remains one of the leading causes of mortality worldwide due to the lateness of diagnosis and lack of access to expert medical treatment. Diagnosis, more so malignancy in CT scans, is not only time-consuming but also vulnerable to human mistakes. This paper presents a blanket AI-powered solution that marries deep learning to make accurate diagnoses of lung cancer with an smart multilingual chatbot to act as a gap-filler between diagnosis and patient understanding. With the Swin Transformer model, the system provides predictions of the risk score of malignancy from the LIDC-IDRI CT scan database. For increased accessibility and usage, Meta's NLLB-200 model is used to translate the medical reports into various languages, with BioBERT supporting natural language question answering from the reports. Ultimately, the objective is to apply early, explainable, and accessible lung cancer screening tools in multilingual patient populations.**

***Index Terms*—Lung cancer detection, CT scan analysis, Swin Transformer, Deep learning, Multilingual chatbot, NLLB-200, BioBERT, Natural language processing, Medical imaging, AI in healthcare, Explainable AI, Streamlit, Patient-centric systems, LIDC-IDRI dataset.**

I. Introduction and Statement of the problem

Lung cancer is the most common cause of cancer-related death globally, with about 1.8 million deaths annually. One of the primary explanations for the poor survival rates is the delayed detection of malignant nodules on CT scans, especially in regions with limited access to healthcare. Additionally, even when diagnosis is available, the technical complexity and language barriers in traditional medical reports complicate patients' ability to understand and participate in treatment decisions.

This project seeks to overcome these challenges by integrating bleeding-edge AI in medical imaging and natural language processing. The platform enables automatic identification of lung cancer markers from CT scan images and interpretation of medical reports in different languages. It also allows patients to chat with a chatbot in answering questions about health and receiving simplified but accurate answers. This intersection of diagnostic accuracy and linguistic access marks a significant leap toward equal, AI-enabled provision of healthcare.

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A. The Rise of AI in Medical Diagnostics

Artificial intelligence has revolutionized diagnostic medicine by offering technology that is able to scan through large amounts of medical information, like high-definition CT scans, quickly and with great accuracy. Models like the Swin Transformer have been found to outperform other models at image classification and are appropriate for interpreting CT scans.

B. Challenges in Patient Communication

Even if proper diagnoses are established, patients tend to struggle with understanding technical medical terminology, especially when it is not in their native language. This leads to confusion, low treatment adherence, and dissatisfaction with care. The issues can be addressed by a multilingual, artificial intelligence-powered chatbot that can interpret medical reports and respond to patient questions in plain language.

C. Project Vision

The goal of the current project is to establish an end-to-end system for lung cancer detection using deep learning along with sophisticated NLP models that encompass multilingual explanation and interactive assistance through a chatbot platform. The system aims to make medical knowledge democratically accessible, expand patient education, and reduce the burden of healthcare.

Problem Statement:

Despite medical diagnosis AI enhancement by technology, there are inherent gaps. They include few real-time, multilingual AI systems to interpret complex medical findings in a comprehensible manner to patients. CT scan interpretation is delayed or incorrect in rural areas due to the fact that there are not sufficient experienced radiologists. Apart from that, medical reports are difficult to comprehend for patients, especially in technical English. It is essential to have an inclusive AI system that not only identifies lung cancer from CT scans accurately but also communicates efficiently through multiple languages.

II. Review of the Literature

A. Deep Learning for Lung Cancer Detection:

Deep learning has shown a lot of promise for lung nodule detection on CT scans. Deep learning's popularity has escalated especially with LIDC-IDRI image dataset utilization to design 3D convolutional neural networks (CNN) for both segmentation and classification tasks. Ironically, it is especially interesting to note that transformer architecture-based such as the Swin Transformer and 3D CNN usage exceeded state-of-the-art machine learning approaches because it allows spatial hierarchy modeling and pattern information from spatial context provided by volumetric input Liu et al., 2021.

Ardila et al., 2019 study demonstrated how Google's model of deep learning matched the performances of experienced radiologists in detecting lung cancer from low-dose CT scans . Similarly, attention-based architectures have further improved lesion localization and malignancy scoring.

B. Transformer-Based Architectures in Medical Imaging:

Transformer models, initially introduced to NLP, now take increasing significance in vision tasks. Swin Transformer, for example, employs hierarchical feature maps and shifted windows to represent images at a large scale. With this, they can make analysis finer using high-resolution CT scans Liu et al., 2021.

Latest studies have shown that Swin Transformer outperforms ResNet and DenseNet in lung nodule classification. Its flexibility and self-attention mechanism can better accommodate the different appearance of pulmonary nodules.

C. Natural Language Processing in Medical Applications:

NLP is crucial for processing and interpretation of clinical text. BioBERT and ClinicalBERT models have been pre-trained with clinical corpora so that they are able to better understand clinical language. Such models enable systems to extract informative information from electronic health records (EHRs), summarize, and respond to domain-level questions Lee et al., 2019.

For instance, BioBERT has demonstrated great accuracy in question answering (QA) and named entity recognition (NER) for biomedical text and thus is a probable choice to be incorporated into conversational medical systems.

D. Multilingual Translation in Healthcare:

To bridge language barriers in healthcare communication, multilingual models like Meta AI's NLLB-200 (No Language Left Behind) have become essential. NLLB-200 supports translation across 200+ languages with high fidelity, particularly for low-resource languages.

Use of such models in healthcare ensures that non-English-speaking patients receive critical medical information in their native language, thereby improving understanding, adherence to medical advice, and overall satisfaction.

E. Conversational AI for Patient Interaction:

Conversational AI technology, in the form of chatbots employing models like GPT-3.5/4, is increasingly being utilized for telemedicine and patient support software. These technologies give patients a natural, interactive way to ask questions and receive reliable answers.

Studies show that AI chatbots improve healthcare access, especially when integrated with medical knowledge bases and personalized context such as past conversations or lab results. When combined with vector-based retrieval and embeddings (such as RAG architecture), these systems can provide accurate, context-dependent responses.

III. Objectives of the Study

The aim of this study is to conceptualize, implement, and deploy an AI-enabled system for early detection of lung cancer from CT scans and presenting results through a multilingual user-friendly interface. Specific goals are as follows:

- Data Acquisition and Preprocessing: Acquire and preprocess CT scan data from publicly available LIDC-IDRI data, including processing DICOM files, detecting nodules, and normalizing images.
- Model Training: To train a Swin Transformer-based deep learning model for lung nodule classification on malignancy scores labeled in the dataset.
- Translation Integration: To use Meta's NLLB-200 model for automatic translation of AI-created medical reports into other native languages and break language barriers for non-native English speakers.
- Chatbot Interface Development: To build a chat interface using Streamlit and integrate BioBERT for answering patient questions in natural language from their medical reports.
- Explainability and Report Generation: Producing human-understandable, clinically relevant reports with explicable explanations of malignancy risk scores, in PDF format and personalized by choice of language.
- Accessibility Focus: To ensure the platform web-based, lean, and deployable in low-resource healthcare environments.

IV. Data Collection

The LIDC-IDRI (Lung Image Database Consortium and Image Database Resource Initiative) dataset The Cancer Imaging Archive, 2015, which is publicly accessible and widely used in medical imaging studies for lung cancer diagnosis, was utilized in this work. As described in Armato et al. Armato III et al., 2011, the LIDC-IDRI dataset provides detailed radiologist annotations on nodules including malignancy scores. Each scan includes DICOM images and XML files containing detailed annotations about pulmonary nodules, including malignancy values ranging from 1 (benign) to 5 (very malignant).

1) Dataset Subset Selection: A subset of 100 patient cases (LIDC-IDRI-0001 to LIDC-IDRI-0100) was selected for initial research and experimentation due to computational constraints and to ensure effective prototyping. This subset was chosen to strike a balance between dataset manageability for model training and evaluation, and maintaining diversity in imaging characteristics.

2) Preprocessing Workflow: Similar to the preprocessing pipeline in the LUNA16 challenge Setio et al., 2017, we converted DICOM files to PNG format, applied histogram equalization, and parsed XML annotations to extract malignancy labels.

TABLE I
DATASET SUMMARY

Attribute	Description
Total Patients	100
Imaging Format	DICOM → PNG (224×224 resolution)
Annotations	Radiologist-labeled XML files containing nodule information and malignancy ratings
Labels	Average malignancy score (range: 1 to 5) computed per patient
Final Dataset	∼18,000+ preprocessed CT slice images, each labeled with malignancy score and nodule count

A Swin Transformer model for malignancy prediction, a ViT model for image classification, and the diagnostic reasoning incorporated into the chatbot interface were all trained using this selected dataset.

V. DATA PREPROCESSING

A systematic data preprocessing workflow was created to prepare the LIDC-IDRI dataset for efficient training and inference. The procedures aimed to convert the unstructured DICOM information into a labeled and organized format suitable for training deep learning models like ViT and Swin Transformer.

1) Converting DICOM to PNG: The original DICOM (`.dcm`) format was used for the raw CT scan images. The `pydicom` and `OpenCV` packages were used to convert these to `.png` images. In order to improve image contrast and the model's visibility of anatomical components, histogram equalization was applied during this step.

2) Normalization and Resizing of Images: To comply with the input size specifications of vision transformer models, each converted PNG image was resized to a fixed resolution of 224×224 pixels. Pixel intensity distributions were normalized using ImageNet mean and standard deviation values.

3) Labeling and Parsing Annotations: Radiologist-assigned lung nodule malignancy scores were extracted from XML annotation files provided with the LIDC-IDRI dataset using the `ElementTree` parser. A single regression label indicating the scan's level of malignancy was calculated for each patient by averaging the malignancy scores of all identified nodules.

4) Filtering Images (CLIP and ViT): To ensure that only relevant lung CT slices were passed to the malignancy prediction model:

- Non-medical or unnecessary images (such as photos or illustrations) were removed using CLIP.
- ViT (Vision Transformer) was trained to differentiate between lung and non-lung CT scans, filtering out irrelevant slices that passed the CLIP stage.

5) Structure of Datasets: Each processed image was associated with its average nodule count and malignancy score, which were obtained from an accompanying Excel sheet. This data was compiled into a structured CSV file named `final_dataset.csv`, with the following columns:

- `image_path`
- `nodule_count`
- `avg_malignancy`

6) Train-Validation-Test Split: The dataset was randomly divided into three subsets:

- 70% training set
- 15% validation set
- 15% testing set

Separate CSV files were generated for each subset, ensuring balanced distribution of malignancy scores across all splits.

VI. EXPLORATORY DATA ANALYSIS (EDA) AND HYPOTHESIS FOR THIS STUDY

The exploratory analysis of the preprocessed CT scan dataset yields valuable new insights into the image properties and malignancy diagnoses. A histogram of nodule counts per patient showed that fewer than six nodules larger than 3 mm were present in most cases. This implies a fair distribution of both high-risk and low-risk examples suitable for supervised learning.

Fig. 1 displays the distribution of nodule counts, while Fig. 2 illustrates the average malignancy scores. Malignancy scores ranged from 0 (no nodules) to 5 (extremely malignant), with most results falling between 1 and 3. This skewed distribution, typical of early-stage cancer screening datasets, supports the model's need to detect subtle pathological features.

To better understand the statistical relationship between nodule count and malignancy, a box plot (Fig. 3) was created. It reveals variations in average malignancy across different nodule counts. While some higher nodule counts are associated with elevated malignancy, a wide variance and presence of outliers (even in low nodule counts) indicate that malignancy cannot be reliably predicted from count alone. This justifies the use of deep learning models capable of identifying more nuanced patterns in imaging data.

Additionally, a correlation heatmap (Fig. 4) showed a moderate positive correlation between the number of nodules and average malignancy score, though not strictly linear. The need for deep learning models that can recognize subtle patterns at the image level beyond basic statistics is thus underscored.

These findings support the use of pre-filtering steps to enhance the quality of input and aid in the creation of a malignancy prediction model that makes use of advanced image transformers.

Based on these insights, the following hypotheses were formulated:

- **H1:** Deep learning models, specifically the Swin Transformer, can accurately predict the malignancy score of a CT scan slice based on image features alone.

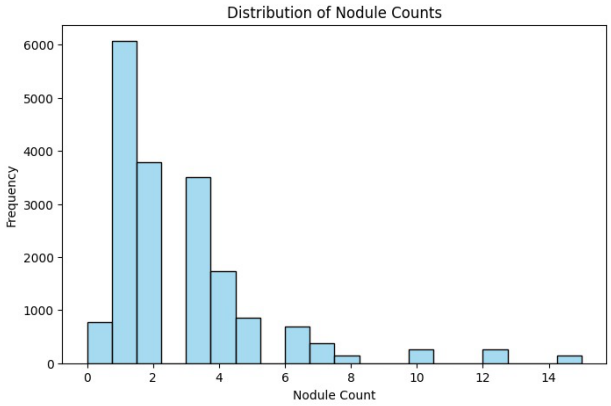

Fig. 1. Distribution of nodule counts per patient

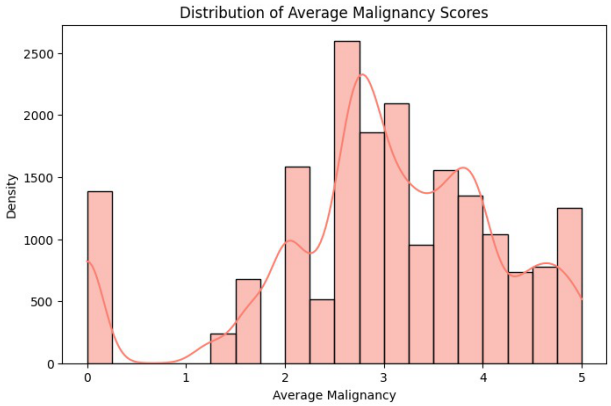

Fig. 2. Density plot of average malignancy scores

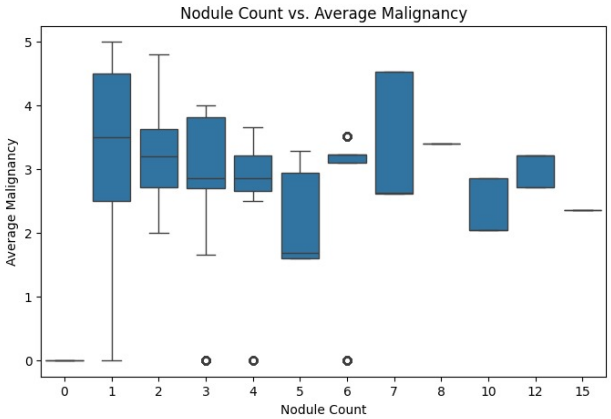

Fig. 3. Box plot showing the relationship between nodule count and average malignancy score

- **H2:** There exists a moderate positive correlation between nodule count and malignancy score, although it is not a sufficient standalone indicator.
- **H3:** Pre-filtering using ViT and CLIP improves the quality of input data, thereby enhancing the accuracy of malignancy predictions.

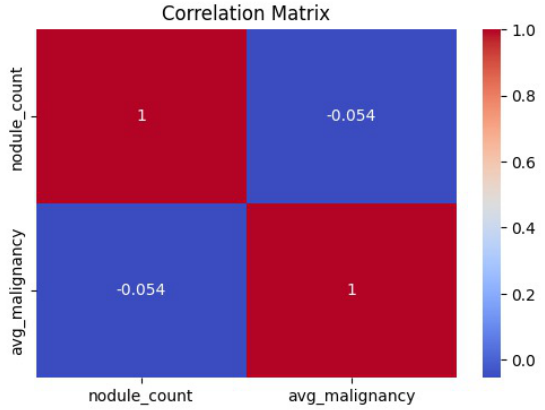

Fig. 4. Correlation matrix of nodule count and malignancy score

- **H4:** Multilingual medical reports and Q&A enabled through NLLB and BioBERT increase patient understanding and accessibility.

VII. Model Description

1) Swin Transformer Model: Swin Tranformer (Shifted Window Tranformer) is powerful type of Vision-Tranformer arcitecture. It uses a shifting method that allows cross-window connections and computes self-attention within local non-overlapping windows in order to process images. Its ability to capture both local and global features (Liu et al., 2021) is made possible by its architecture, which is essential for observing complex patterns in medical imaging such as CT scans.

In order to perform regression on CT scan slices from the LIDC-IDRI dataset, we modified the Swin-Tiny variation for our project. ImageNet data were used to normalize each input image and reduce it to 224 x 224. A continuous malignancy score that indicates the probability of lung cancer existence is the final prediction result. With a learning rate of 1e-4, we used the Adam optimizer to optimize the model after training it with Mean Squared Error (MSE) loss. With an MSE of 0.0192, MAE of 0.0500, and R2 score of 0.9872, the model demonstrated outstanding performance metrics and the capacity to predict malignancy with high accuracy.

We obtained efficient and clinically applicable AI-powered lung cancer diagnosis that is both reliable and understandable by including Swin Transformer into our multilingual chatbot workflow.

2) NLLB: A modern multilingual translation system called Meta's No Language Left Behind (NLLB) model was created to overcome the drawbacks of traditional machine translation, particularly when dealing with low-resource languages. With a single, integrated model design, NLLB allows high-quality translation in more than 200 languages, many of which are underdeveloped. By translating medical information and chatbot conversations into the user's choice language, NLLB helps contribute to our project's goal of making the lung cancer detection chatbot accessible to a wide range of

users. This provides equality and eliminates barriers to health communication by enabling patients from different linguistic origins to access and comprehend important health information in the language they are most familiar with (AI, 2022).

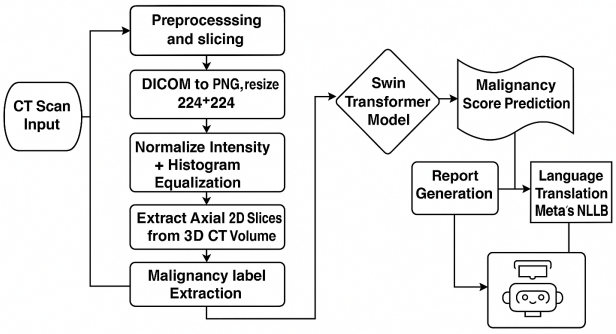

Fig. 5. System architecture showing the preprocessing, Swin Transformer-based malignancy prediction, multilingual translation with NLLB, and chatbot integration.

3) CLIP and ViT Models:

a) CLIP: Contrastive Language–Image Pretraining (CLIP), developed by OpenAI, is a vision-language model that learns to project images onto natural language captions through contrastive learning. In our project, we used CLIP for image validation. It checks whether the uploaded scan is a medical CT scan or not before proceeding to the next step. CLIP works by embedding text and images into a shared space and measuring similarity scores between them. We utilize this by comparing user-submitted images to the text query "lung CT scan." If the similarity score falls below a predefined threshold, the image is classified as a non-lung CT scan. This process helps reject incorrect inputs—such as X-rays or ordinary photographs—and reduces the risk of false predictions Radford et al., 2021.

b) : ViT

Vision Transformer (ViT) is a transformer-based deep learning model that applies mechanisms originally designed for natural language processing to visual data. In our project, CLIP is teamed with ViT to enhance image verification. ViT processes the global context of an image and helps confirm whether the uploaded data is a valid CT scan or just irrelevant content. It converts the CT scan into feature embeddings, which are cross-validated against recognized lung CT patterns for authenticity. The model is pre-trained on a large dataset and fine-tuned specifically to identify lung-related structures Dosovitskiy et al., 2021.

By combining CLIP's multimodal reasoning with ViT's spatial understanding, the model achieves high validation accuracy before proceeding to malignancy analysis using the Swin Transformer.

4) BioBERT: WE chose BioBERT a biomedical domain version of BERT to for the questions and answers section of our chatbot. Trained on large biomedical corpora (PMC papers and PubMed abstracts), BioBERT enables rich contextual understanding of complex medical terminology and its interactions. In our project, we used BioBERT for QnA Patients can ask questions, such as What is the malignancy score and What does this report tell us about the lung cancer and something like that. BioBERT interprets the questions and give user-friendly answers. This step mainly helps to bridge the gap between patients and clinical findings of the report Lee et al., 2020.

5) Chatbot Interface: The user-facing feature of the lung cancer detection system is the chatbot interface, which was constructed with Streamlit. With a simple dropdown menu, users can submit CT scan photos and choose their chosen language. After upload, a picture is verified as a valid lung CT scan using the CLIP and ViT models. The Swin Transformer model then receives the verified image and predicts a malignancy score. This prediction is used to create a patient-friendly medical report, which Meta's NLLB model then translates into the user's preferred language. In addition, the chatbot has BioBERT, which allows users to ask follow-up questions about their medical report and get insightful, medically based answers. The chatbot is a thorough, user-friendly, and interactive tool for AI-assisted medical reporting, and the final report is available for download as a multilingual PDF.

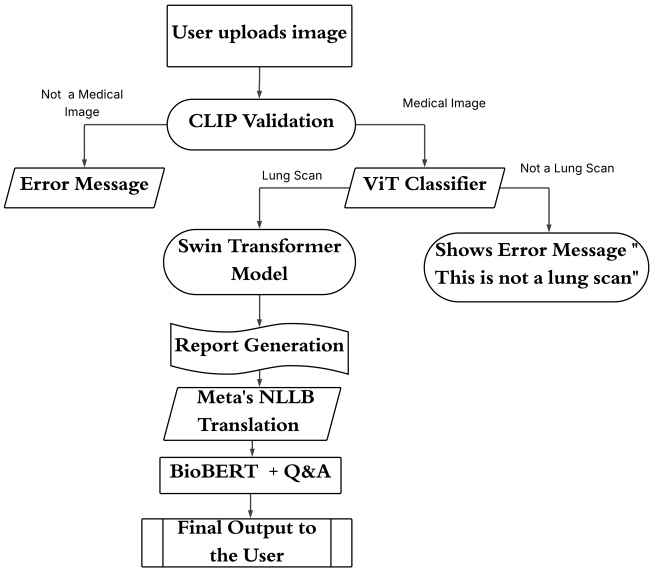

Fig. 6. Chatbot interface flow showing user interaction...

VIII. Data Visualization and Results Report

1) Model Performance (Swin Transformer): Using the validation dataset, the Swin Transformer model showed excellent predictive accuracy. The model's R^2 score of 0.9872, Mean Squared Error (MSE) of 0.0192, and Mean Absolute Error (MAE) of 0.0500 demonstrate its ability to accurately predict malignancy scores from CT scan slices.

The following histogram illustrates how close most predicted malignancy scores are to actual values, indicating the model is well-trained with minimal bias.

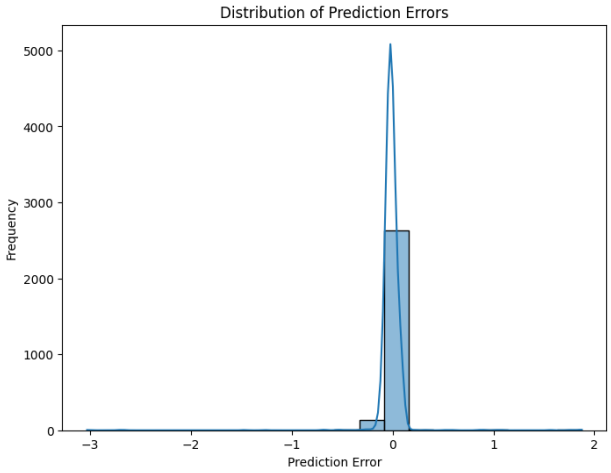

Fig. 7. Histogram of prediction errors showing that most predicted malignancy scores are close to actual values, indicating minimal bias and a well-trained model.

The scatter plot below further confirms the model's regression capability. Most predicted scores are closely aligned with actual values along the diagonal line, showing strong correlation and minimal deviation.

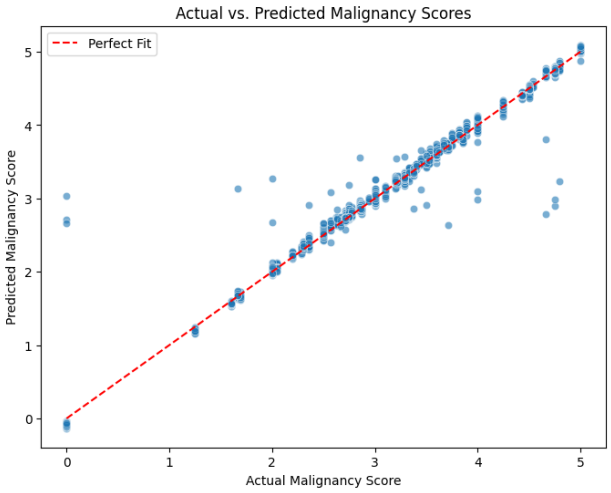

Fig. 8. Scatter plot comparing predicted malignancy scores with actual labels. Points closely aligned along the diagonal indicate strong prediction accuracy.

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2) Chatbot Interface Results: The following screenshots showcase the functionality and workflow of the multilingual chatbot system for lung cancer detection. It highlights the user experience from image upload to final report generation, along with real-time interaction and language accessibility features.

Step 1: Uploading the CT Scan

Users begin by uploading their CT scan image through an intuitive drag-and-drop interface. The system supports common formats like JPG, JPEG, and PNG, up to 200MB.

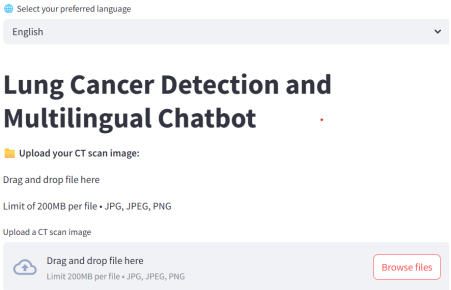

Fig. 9. Chatbot interface prompting the user to upload a CT scan.

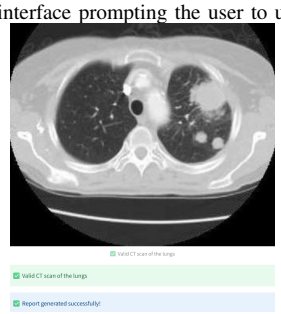

Fig. 10. Feedback confirming the uploaded CT scan is valid.

Step 2: AI-Generated Medical Report

Once the scan is validated, the Swin Transformer model analyzes it and produces a detailed, patient-friendly medical report. The report includes a malignancy risk score and associated recommendations.

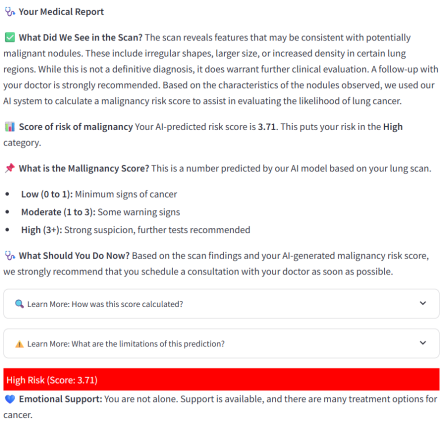

Fig. 11. Auto-generated medical report with malignancy score (3.71) and interpretation. The risk category is highlighted for user clarity.

Step 3: Language Selection Interface

To improve accessibility, the chatbot includes a dropdown menu where users can choose from over 50 languages. This ensures non-English speakers can interact comfortably.

Step 4: Translated Medical Report

Using Meta's NLLB-200 model, the chatbot translates the medical report into the user's selected language, ensuring comprehension across diverse linguistic backgrounds.

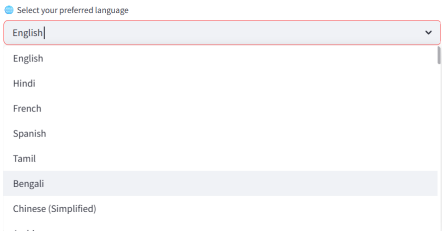

Fig. 12. Language selection dropdown list allowing users to choose from over 50 languages. This enables full translation of chatbot responses and medical reports.

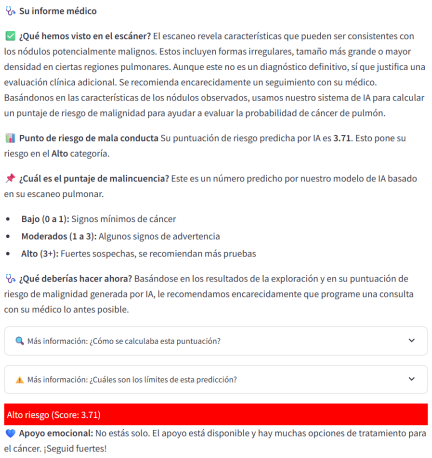

Fig. 9. The medical report translated to Spanish using Meta's NLLB-200 model, showcasing multilingual accessibility.

Step 5: Medical Question Answering

The chatbot allows users to ask questions about their report in natural language. BioBERT interprets these queries and returns easy-to-understand, clinically accurate answers.

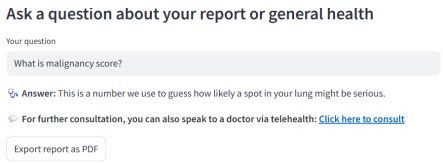

Fig. 13. Question-answering module using BioBERT for medical concepts. Users can ask about scores and receive simplified explanations.

IX. Conclusion

This work succeeded in well-designing an AI-based multilingual chatbot system for early detection of lung cancer, highly enhancing patient accessibility and readability. Adopting cutting-edge deep learning techniques, the Swin Transformer performed well to predict malignancy scores from CT scans with robust performance (MSE: 0.0192, MAE: 0.0500, R^2: 0.9872). To make it reliable, the incorporation of CLIP and ViT models ensured natural validation of CT scans, excluding unwanted or non-lung images with high efficiency. Additionally, multilingual capabilities were integrated flawlessly with Meta's NLLB translation model, providing medical reports and chatbot conversations in over 50 languages. BioBERT provided accurate, user-friendly answers to patient questions, translating complex medical terms into understandable language for patients.

In summary, this system fills current gaps through the combination of accurate medical diagnostics, multilingual patient-friendly communication, and interactive user engagement. It is highly potential for deployment in global healthcare settings, particularly low-resource or multilingual contexts, and thus significantly contributes to improved lung cancer screening and patient outcomes. Future extension may involve scaling this method to other medical conditions, model explainability, and further clinical validations to further solidify the effectiveness and reliability of the chatbot in real-world contexts.

X. Acknowledgment

We would like to extend our appreciation to the developers and contributors of the publicly available LIDC-IDRI dataset, whose detailed annotation work enabled us to have strong training and validation of our models. We appreciate Meta AI for developing the NLLB translation system, which significantly increased the availability of our multilingual chatbot. Also, we would like to thank the authors of BioBERT for having provided a strong NLP model that enhanced our system's capability to offer accurate medical information interactively. Finally, we thank our academic institution and advisors for their constant support and precious guidance throughout this research.

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